(Python-Day-10)

sort() :-

→ sort is list method. which is used change the order of elements

Syntax → it might be Asc or Desc

list.sort()

→ Assc → small to Big
Desc → Big to small

sort → Small to Big
Big + small

a = [30,10,20,50,40]
a.sort()
print(a)
a = [10,20,30,40,50]

a = ['q','J','S','P','A']

a.sort()
print(a) # ['A','q','J','P','S')

a = ["akhi", 'abhi"]
a.sort()
print(a) ['abhi', 'akhi')

a = [1,"Jagadesh",1.0,True, ~~com~~ 2+3j]
a =sort()
print()

→ Error → TypeError →

Desc :-

a = [40,30,50,10,20]

a.sort(reverse=True)

print(a)

False → True

Big :- 50,40,30,20,10

sort → using function

```
{ def myfun(a):
      return len(a) } → Block code
```

```
def myfun(a):
    return len(a)
```

b = ['Surendra','Jaya','gopi','prasadh','Akhi']

b.sort(key=myfun())

print(b)

a=Surendra

len(a) # 8

4

4

7

5

Asce → small

4,4,5,7,8

→ memory implementation :- Timsort → python.exe → ~~Best~~ machine code

python - sort

Obj ID

[10,20,30,40

[20,~~3~~40,30,10]

20 30 40

20 30 10 40

Sort

asc → Dec

Small to Big's

syntax list.reverse()

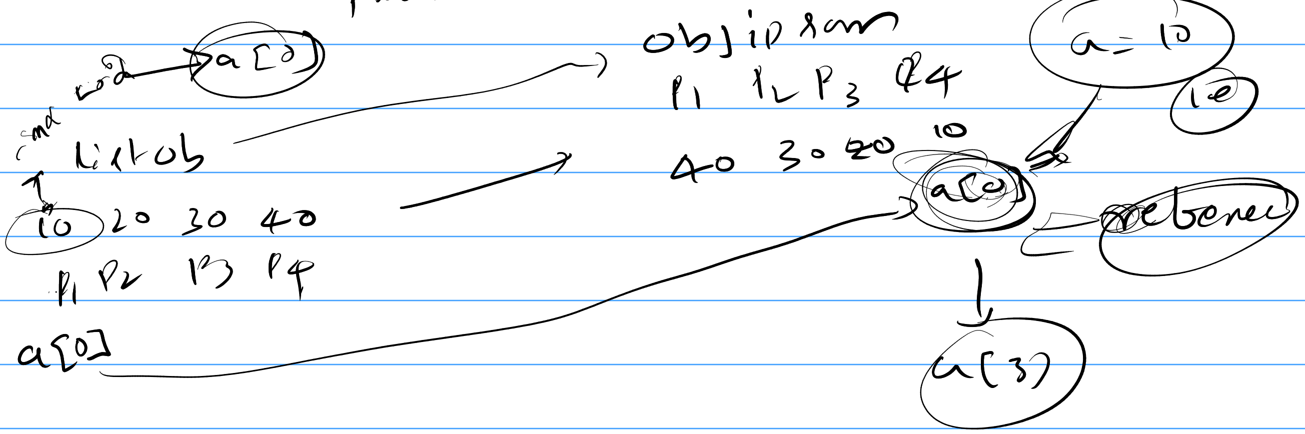
reverse: → → asc, Des → Imp

a = [10, 30, 4, 20]

a = [20, 40, 30, 10]

print(a[::-1]) →

print(a.reverse()) →


2nd ind
a[0]
obj id sam
P1 P2 P3 P4
40 30 20 10
a = 10
10
list ob
10 20 30 40
P1 P2 P3 P4
a[0]
a[0]
reference
a[3]

copy() : copy is list method which can created a newlist using old list

~~lis~~ new_list = old_list.copy()

↳ print(new_list)

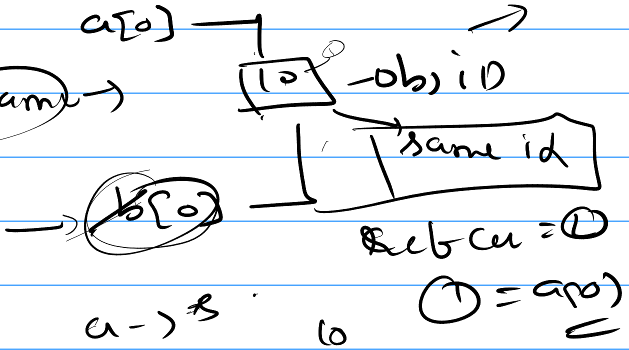
a = [10, 20, 30]

→ b = a.copy() →

print(a) # [10, 20, 30] → 1000

print(b) # [10, 20, 30] → 2000

print(id(a[0]))
print(id(b[0])) } same →


a[0]
10
obj iD
same id
b[0]
Refcou = ①
① = a[0]
a → 3
10

Join list:

→ extend() ✓

→ +

a.extend(b) = [1,2,3,4]

a = [1,2]

b = [3,4]

c = a+b

print(c)

[1,2,3,4]

1000 2000

1 2

a = 2

3000 4000

3, 4

b = 2

4

|1000| 2000 |3000| 4000|

c = [10 20 30 40]

print(c) = 1

Tuple:

→ 1. Built in data structure → ( ) → python.exe

↓

<class Tuple>

→ immutable

↳ (0) methods →

→ ordered: inside element not changed in every time execution

→ Duplicate values support

a = ('Jagadi', 'Jagadi', 10, 20)

→ slicing, indexing support → obj id →
↳ cannot be modified

→ single → multiple values →
→ values cannot be change

a = (10, 20, 30, 40)

→ Tuple is built in python data structure.

why do tuple exist:

→ 1000 → a = ( ) → operation performs

still speed

Hashable → immutable

UnHashable → mutable

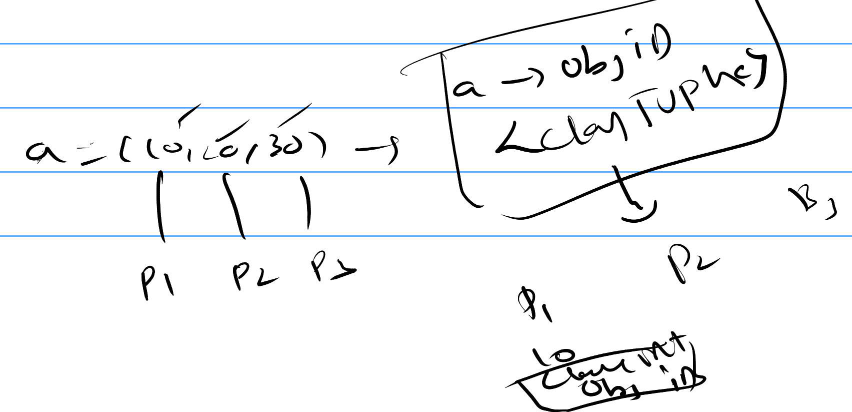
→ tuple len()

→ using the len()

a = (10, 20, 30)

print(len(a)) # 3

memory exp


a = (10, 20, 30) →
P1
P2
P3
a → obj id
<class Tuple>
P1
P2
P3
10
cannot obj id